

Matthew De Jesus

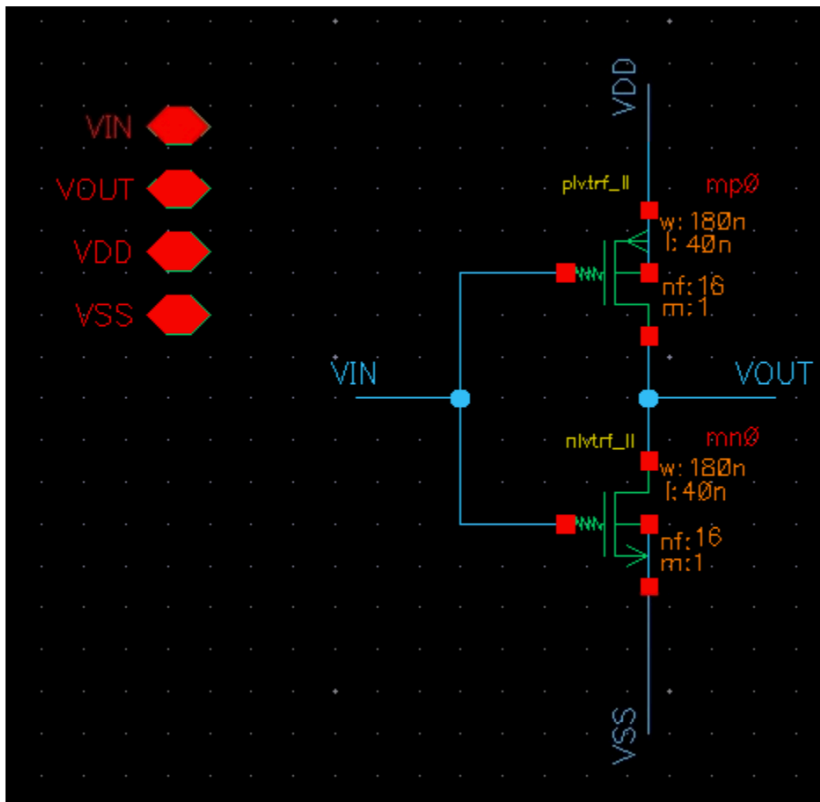
May 12th 2025

6.2080 Semiconductors

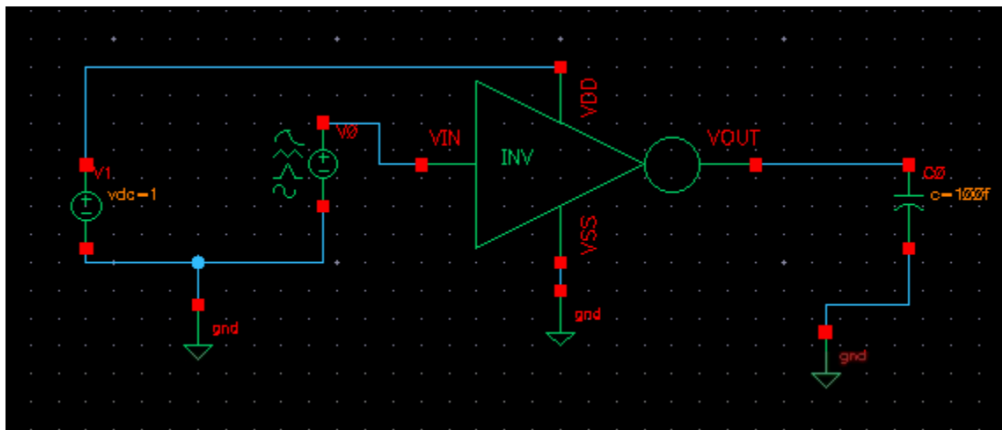
Final Project

Inverter

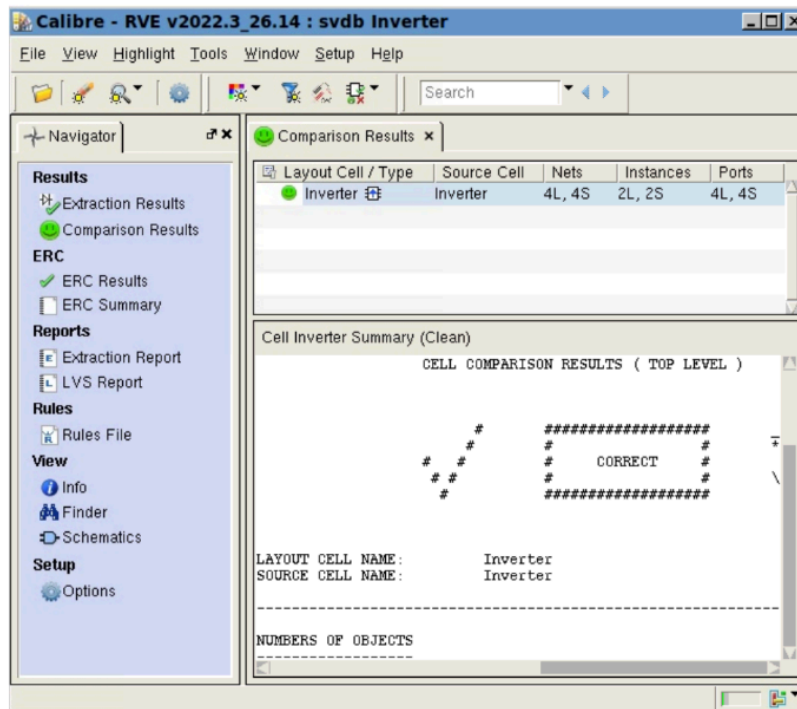
Inverter Schematic



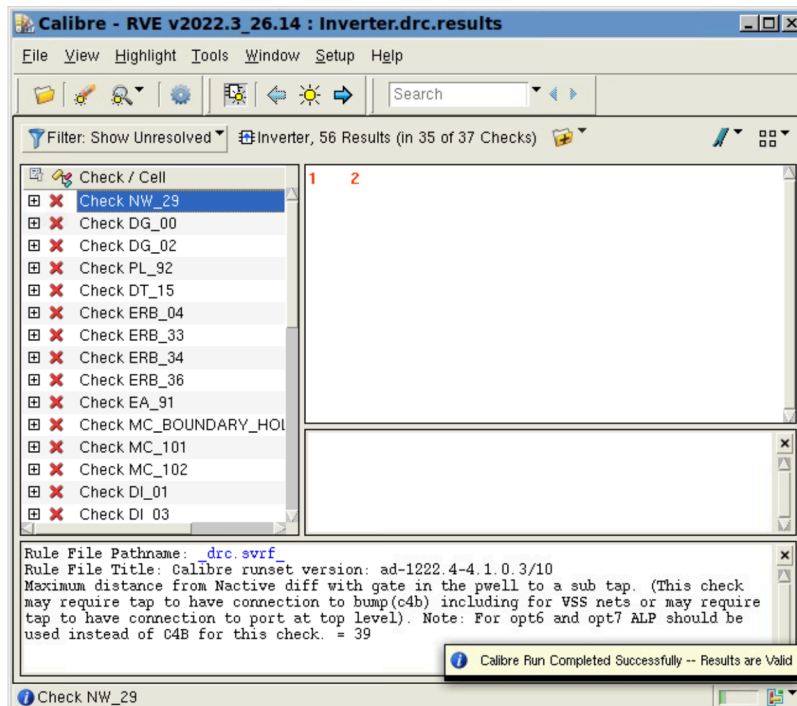
Test Bench and Symbol



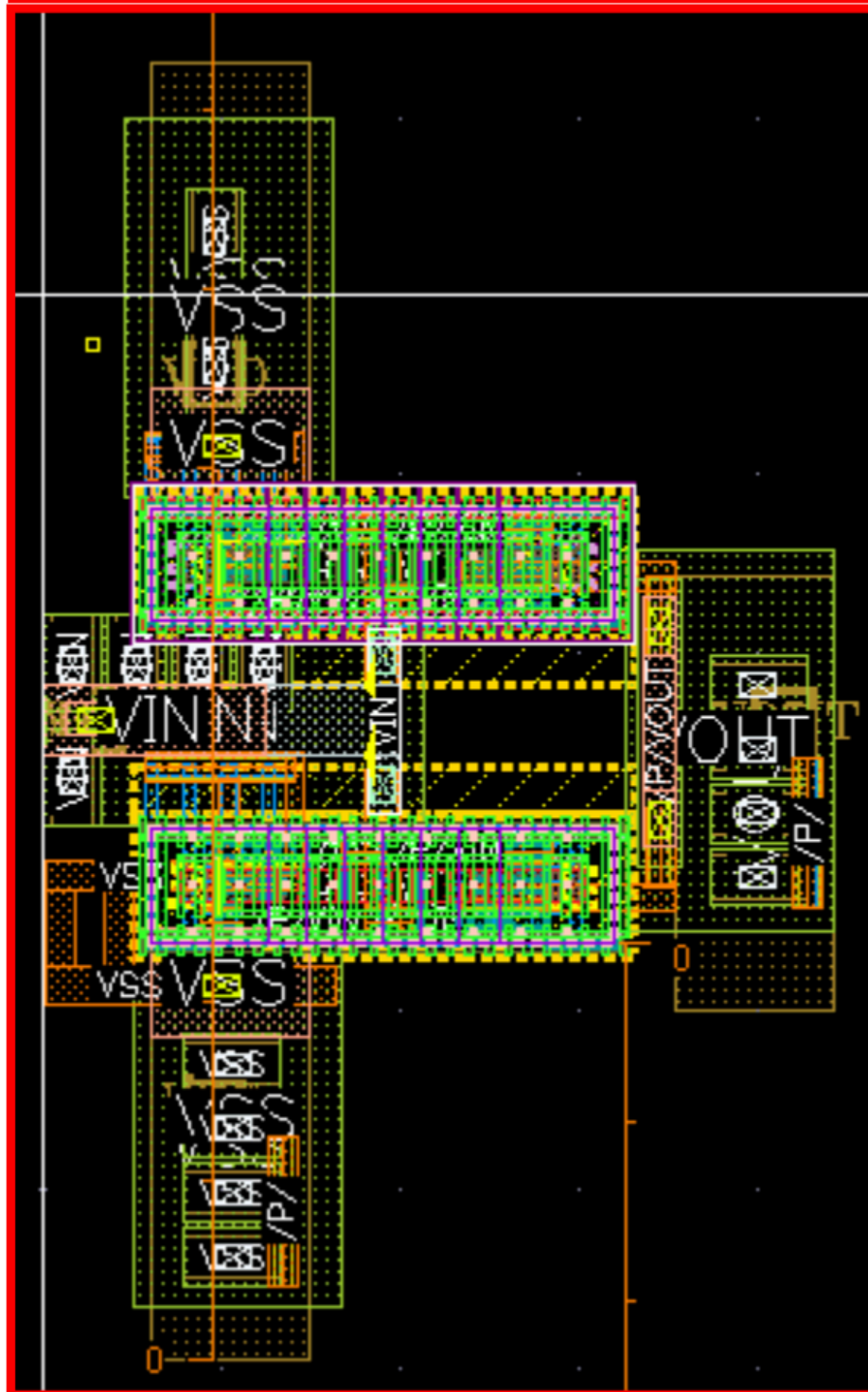
Inverter LVS Clean



Inverter DRC Clean

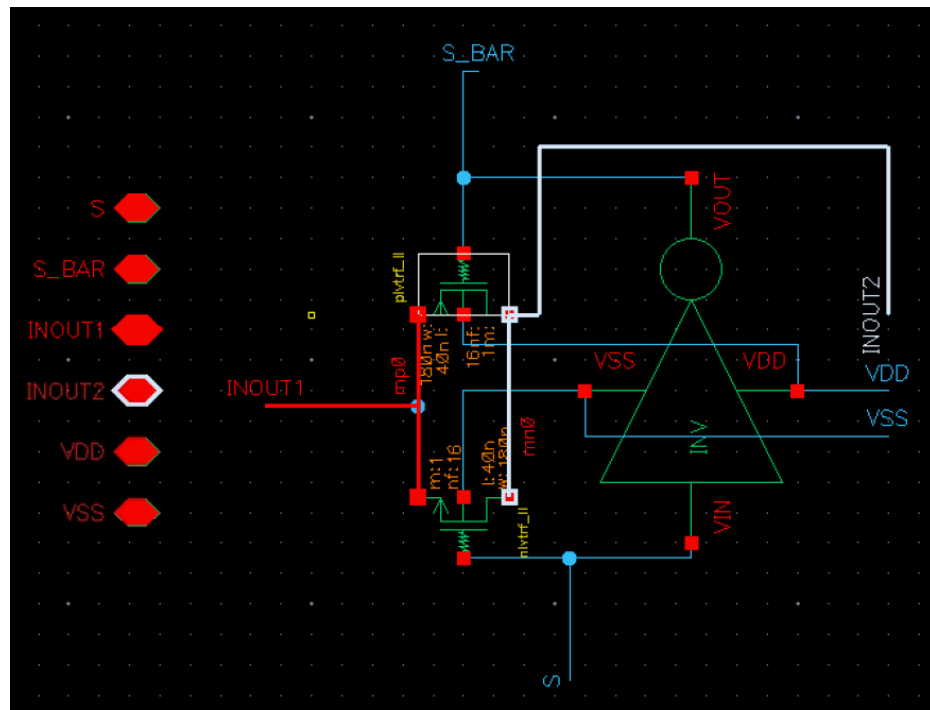


Inverter Layout

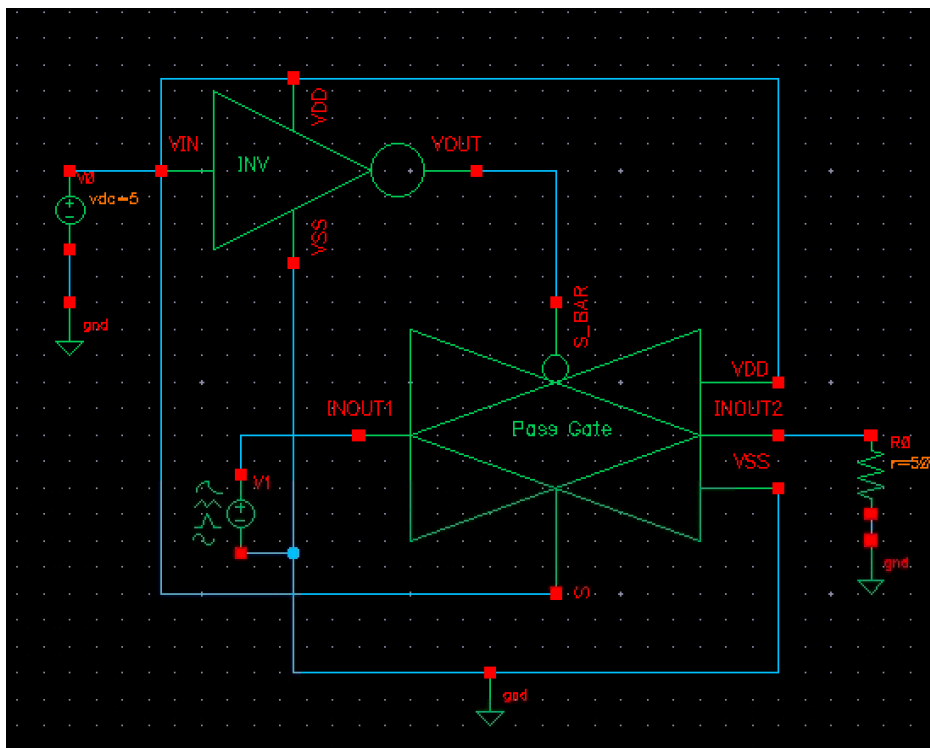


Pass Gate

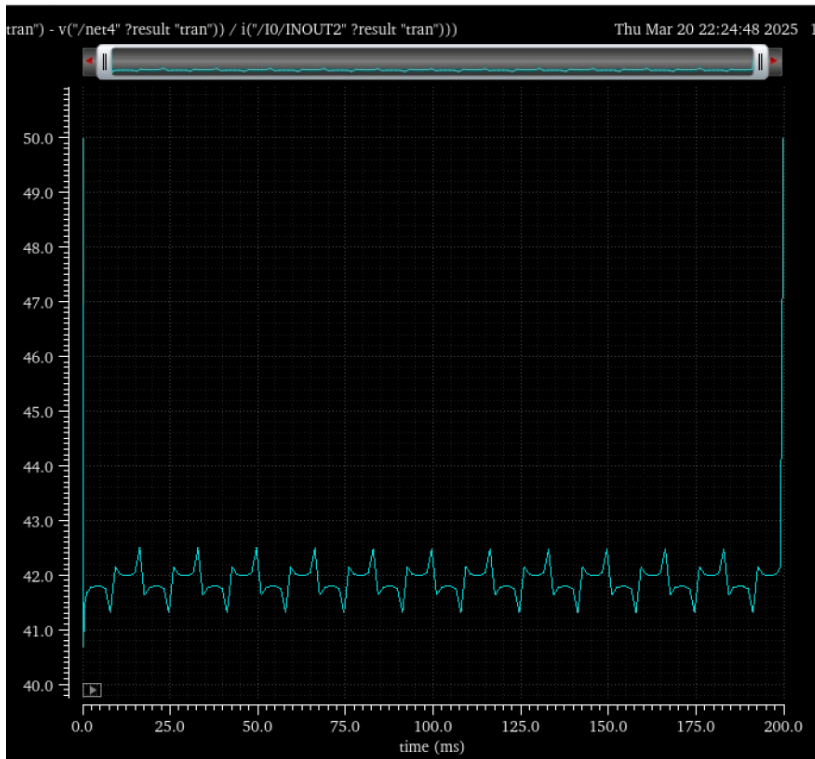
PassGate Schematic



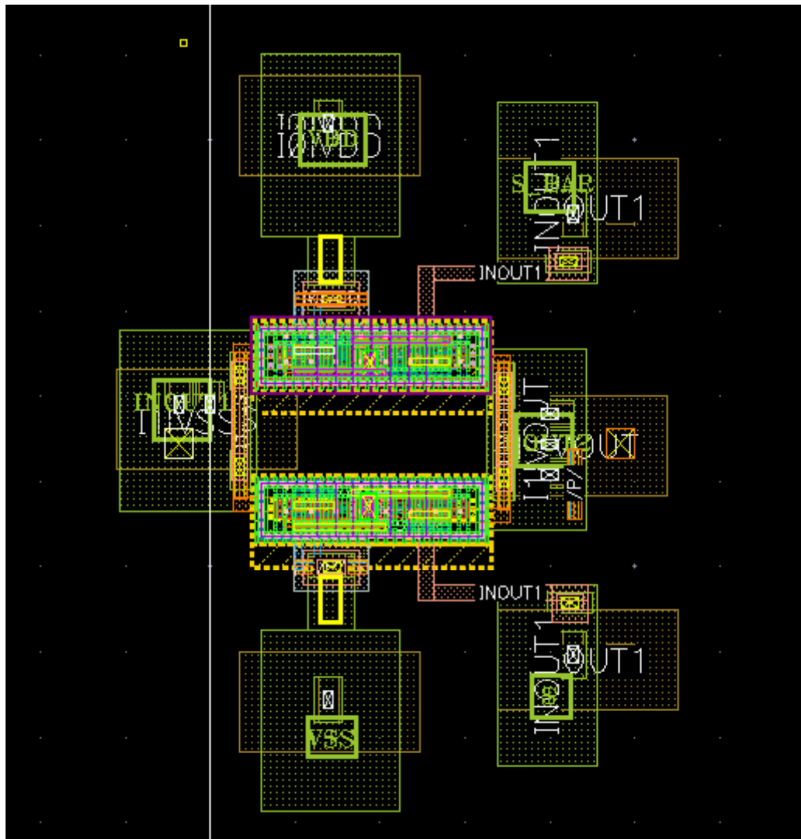
PassGate Test Bench



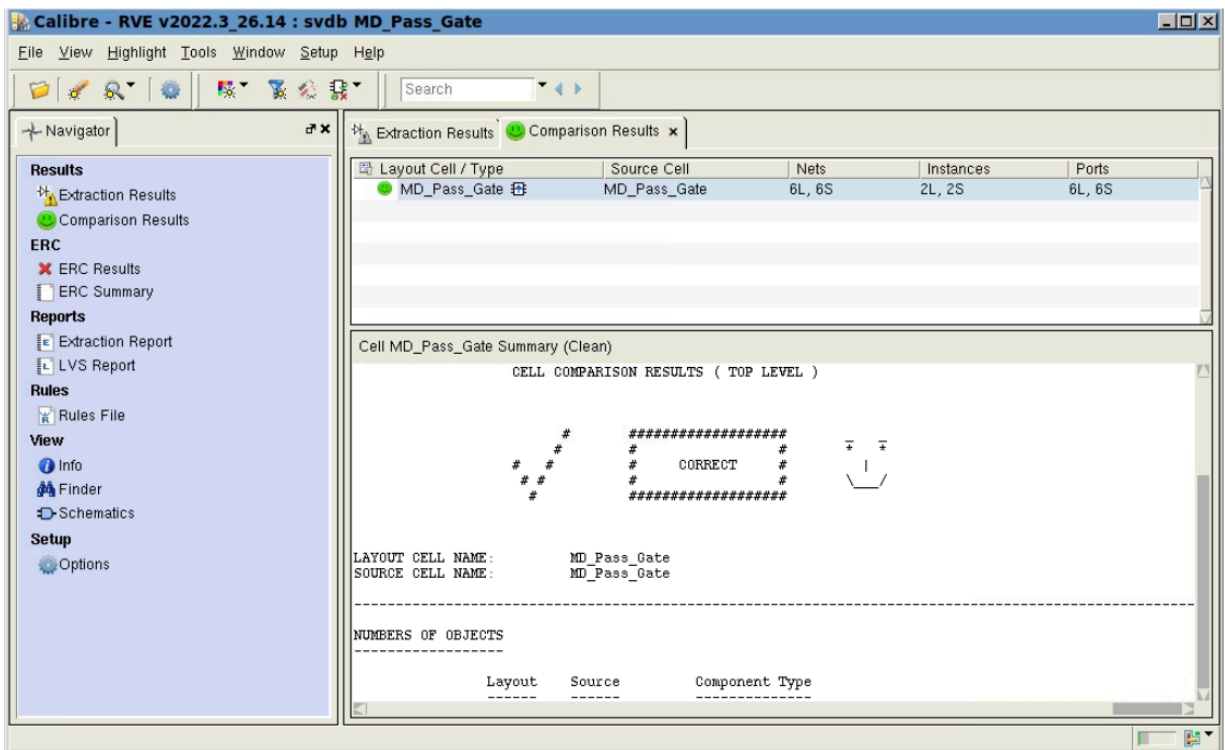
PassGate Resistance



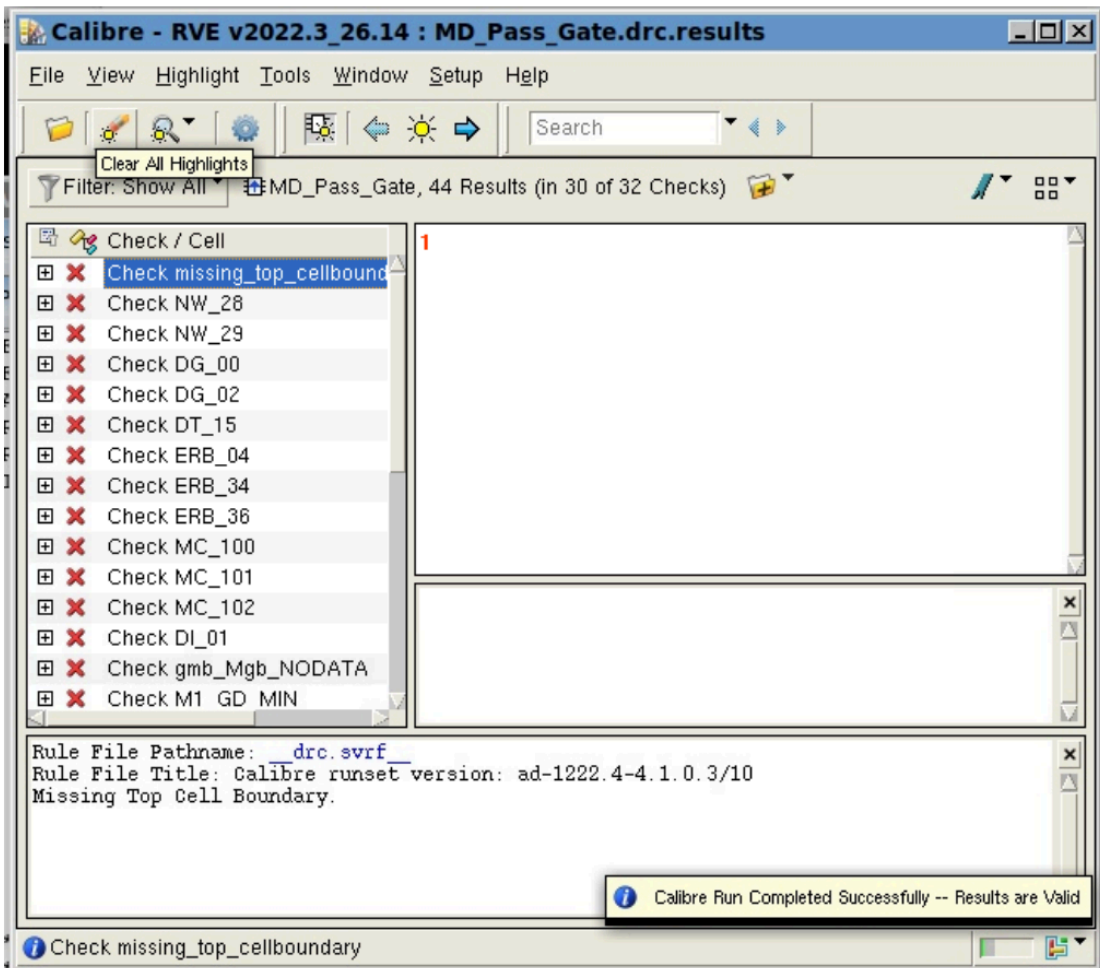
Pass Gate Layout



LVS Clean

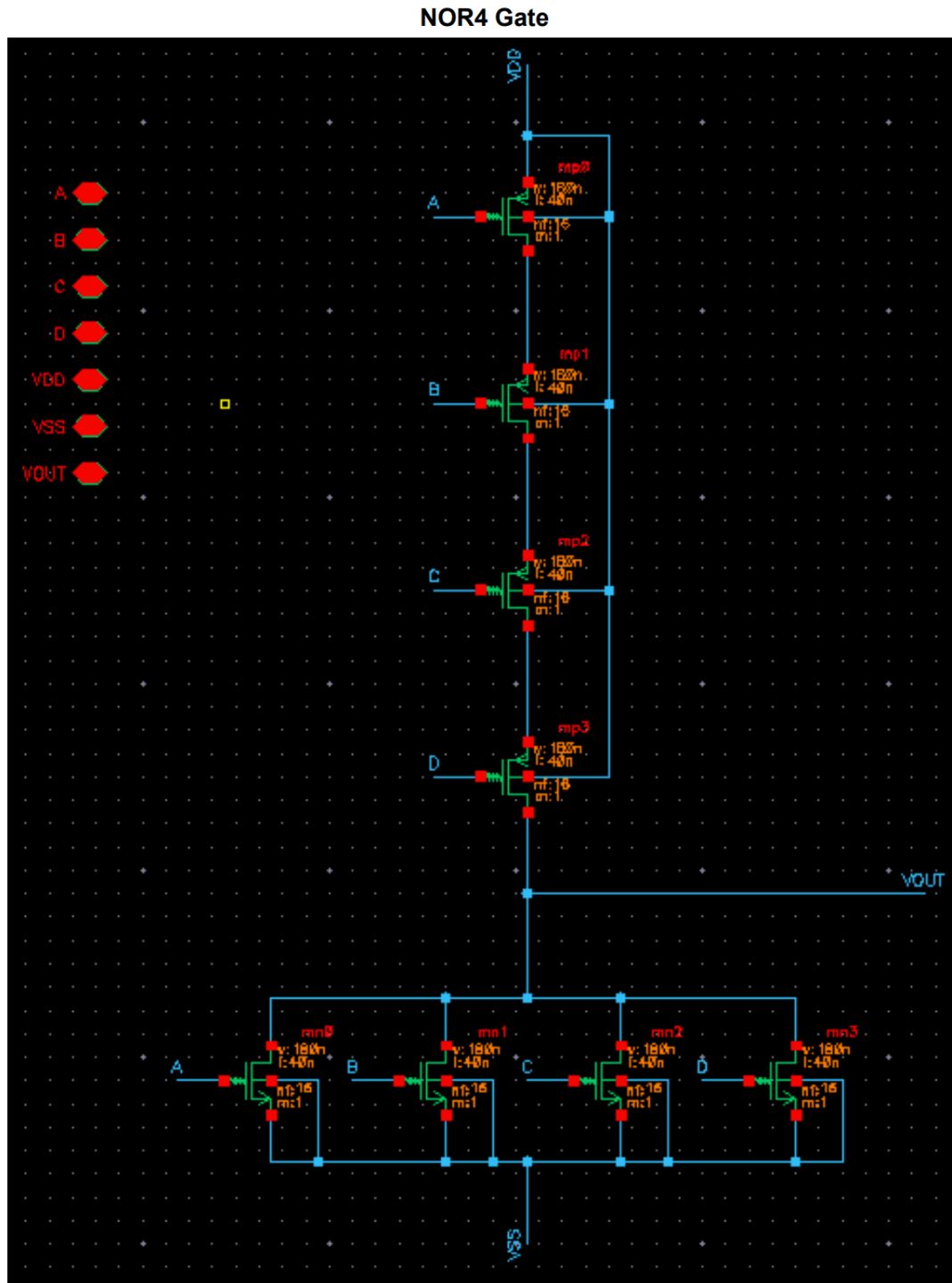


DRC Results

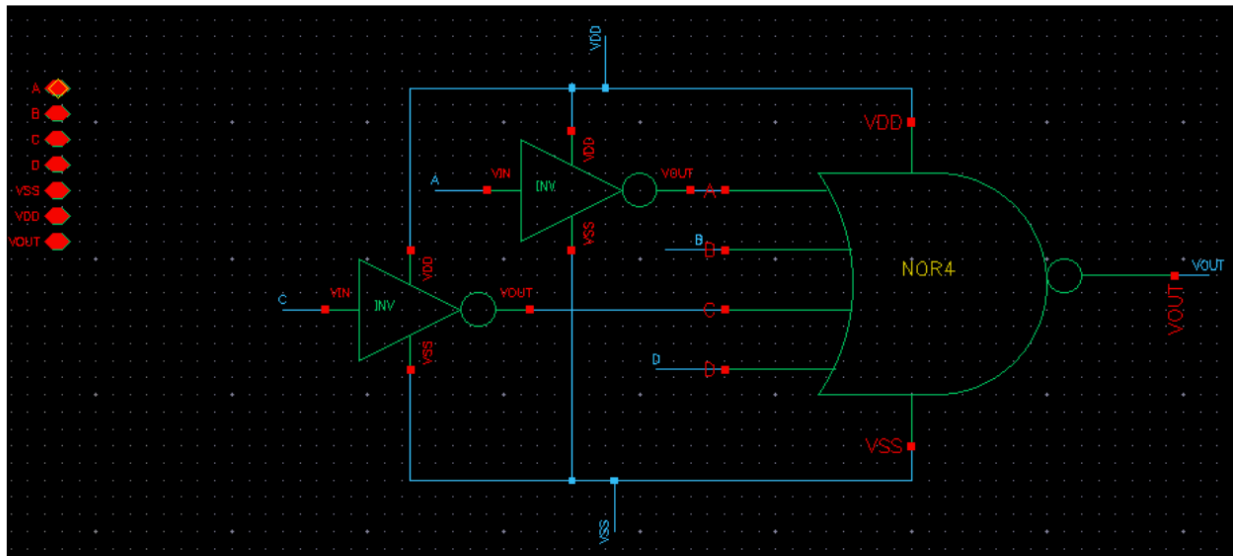


Code Checker

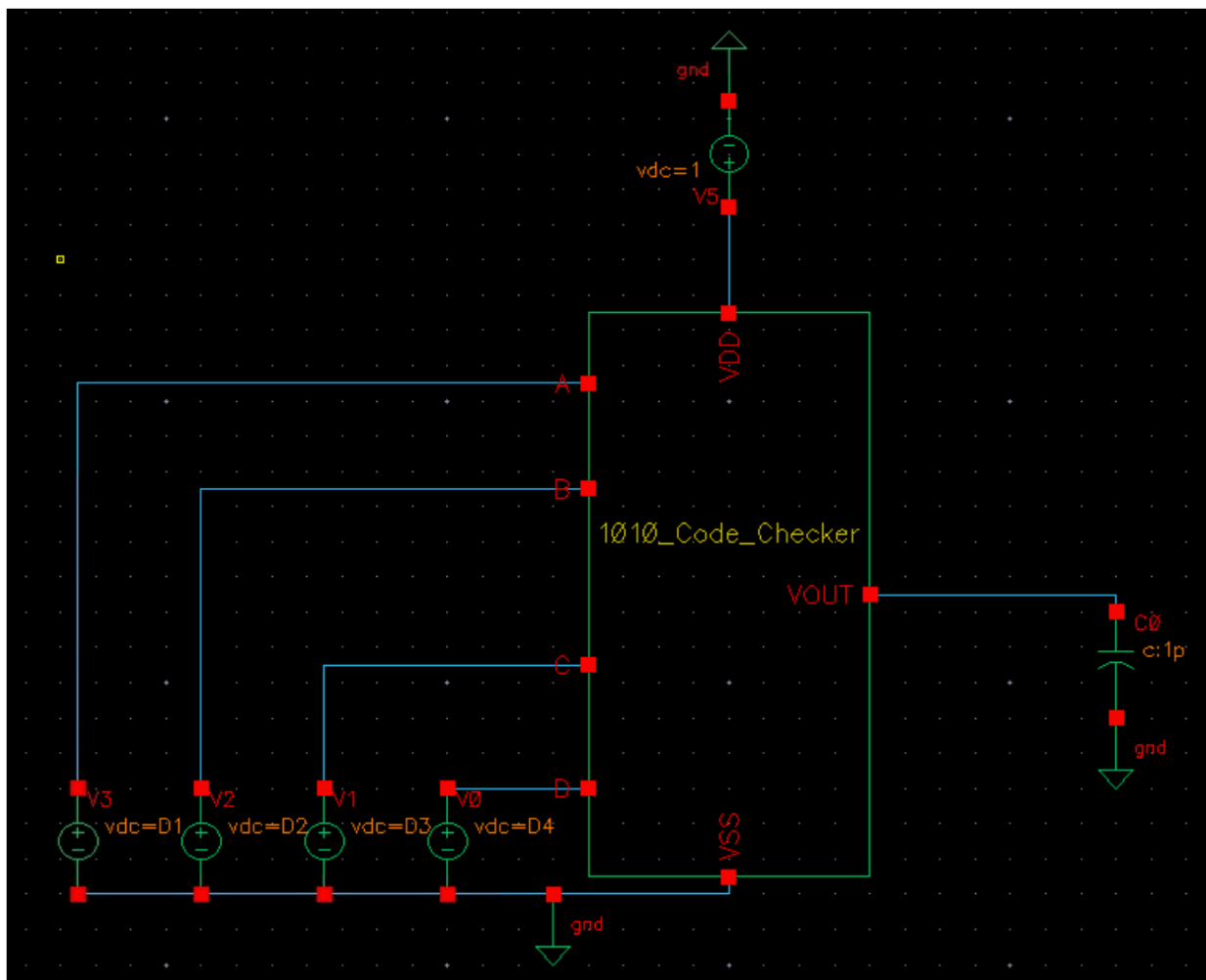
Code Checker NOR4 Schematic



Code Checker



Test Bench



Code Checker Test Bench

Test Bench Results

16/32 rows

Point	Test	Output	Nominal	Spec	Weight	Pass/Fail
Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	PSET_4_MD_Cod...	VDC("/net7")	1.029u			
Parameters: D' =0, D2=0, D3=0, D4=1						
2	PSET_4_MD_Cod...	VDC("/net7")	424.8n			
Parameters: D' =0, D2=0, D3=1, D4=0						
3	PSET_4_MD_Cod...	VDC("/net7")	6.032u			
Parameters: D' =0, D2=0, D3=1, D4=1						
4	PSET_4_MD_Cod...	VDC("/net7")	1.032u			
Parameters: D' =0, D2=1, D3=0, D4=0						
5	PSET_4_MD_Cod...	VDC("/net7")	424.3n			
Parameters: D' =0, D2=1, D3=0, D4=1						
6	PSET_4_MD_Cod...	VDC("/net7")	236.5n			
Parameters: D' =0, D2=1, D3=1, D4=0						
7	PSET_4_MD_Cod...	VDC("/net7")	1.027u			
Parameters: D' =0, D2=1, D3=1, D4=1						
8	PSET_4_MD_Cod...	VDC("/net7")	424.5n			
Parameters: D' =1, D2=0, D3=0, D4=0						
9	PSET_4_MD_Cod...	VDC("/net7")	6.149u			
Parameters: D' =1, D2=0, D3=0, D4=1						
10	PSET_4_MD_Cod...	VDC("/net7")	1.033u			
Parameters: D' =1, D2=0, D3=1, D4=0						
11	PSET_4_MD_Cod...	VDC("/net7")	999.5m			
Parameters: D' =1, D2=0, D3=1, D4=1						
12	PSET_4_MD_Cod...	VDC("/net7")	6.306u			
Parameters: D' =1, D2=1, D3=0, D4=0						
13	PSET_4_MD_Cod...	VDC("/net7")	1.029u			
Parameters: D' =1, D2=1, D3=0, D4=1						
14	PSET_4_MD_Cod...	VDC("/net7")	424.9n			
Parameters: D' =1, D2=1, D3=1, D4=0						
15	PSET_4_MD_Cod...	VDC("/net7")	6.072u			
Parameters: D' =1, D2=1, D3=1, D4=1						
16	PSET_4_MD_Cod...	VDC("/net7")	1.032u			

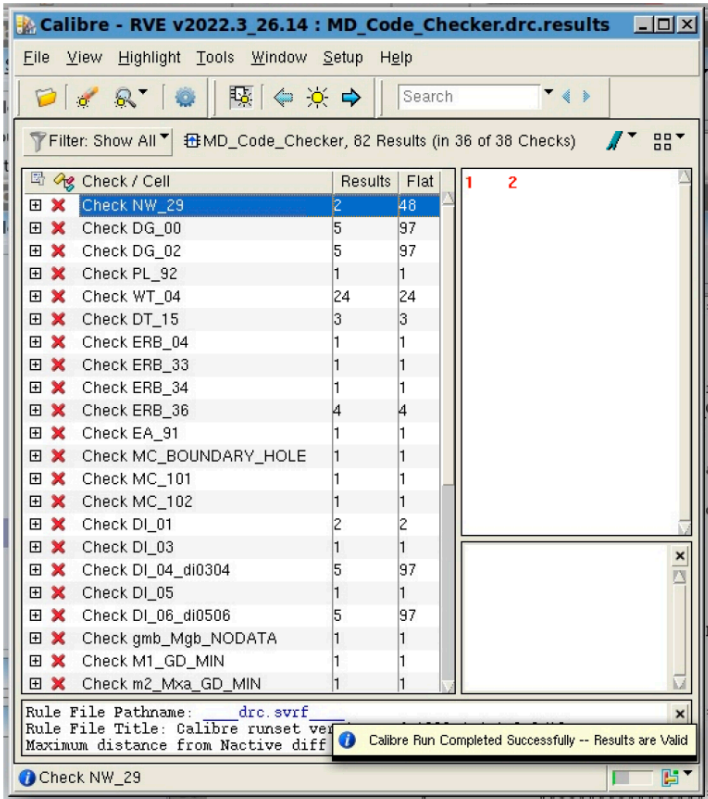
Code Checker Layout

Layout

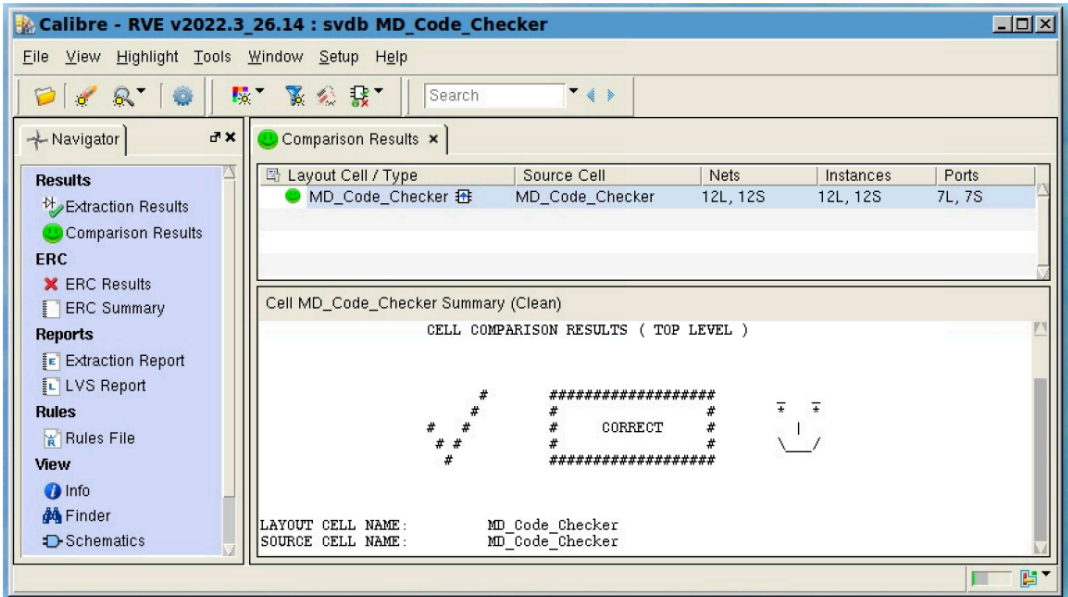


Code Checker DRC and LVS

DRC Check

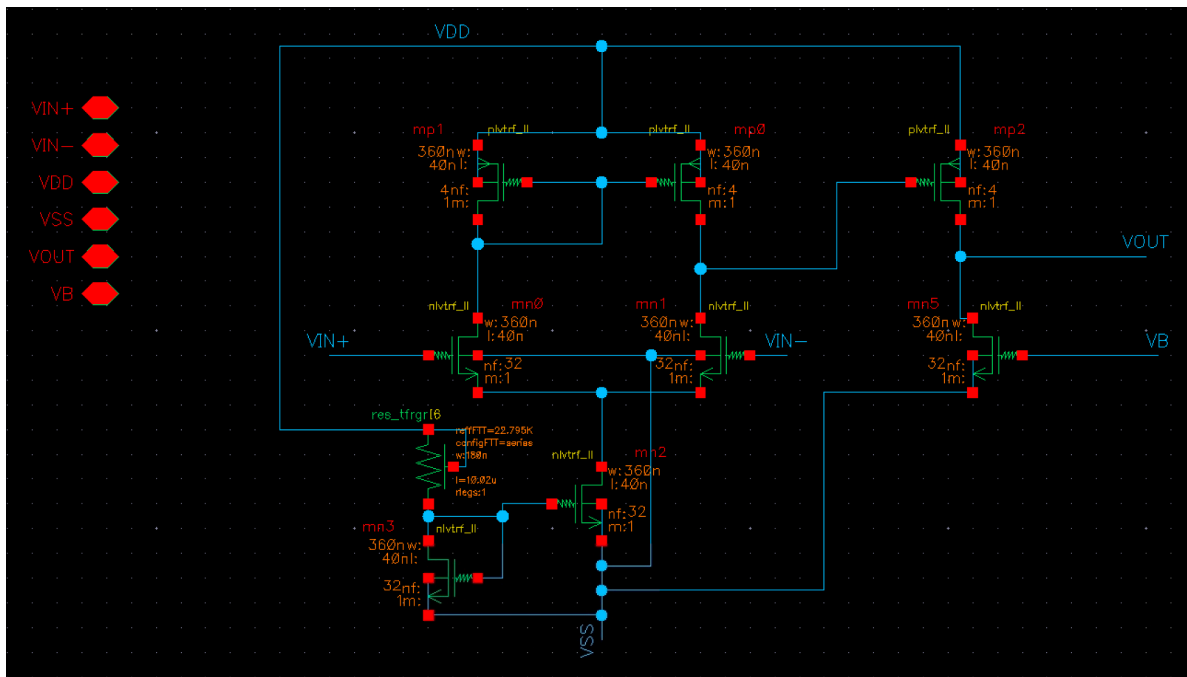


LVS Clean

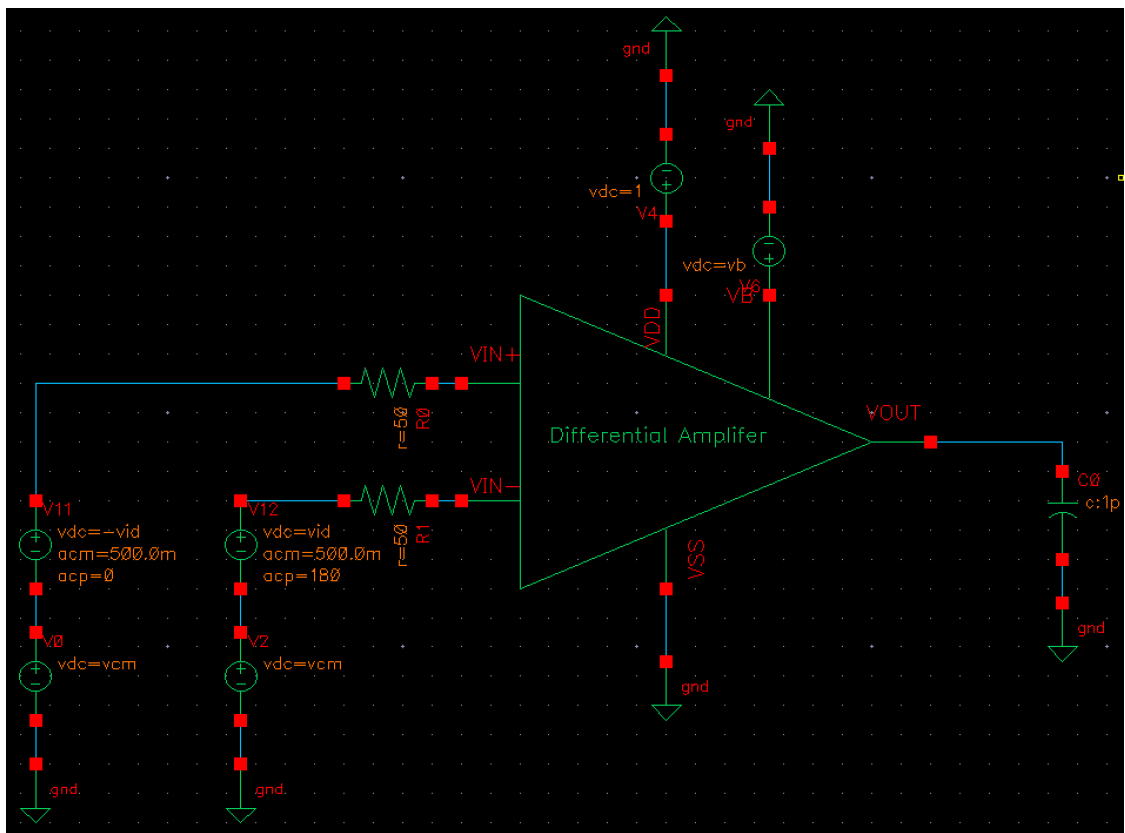


Amplifier

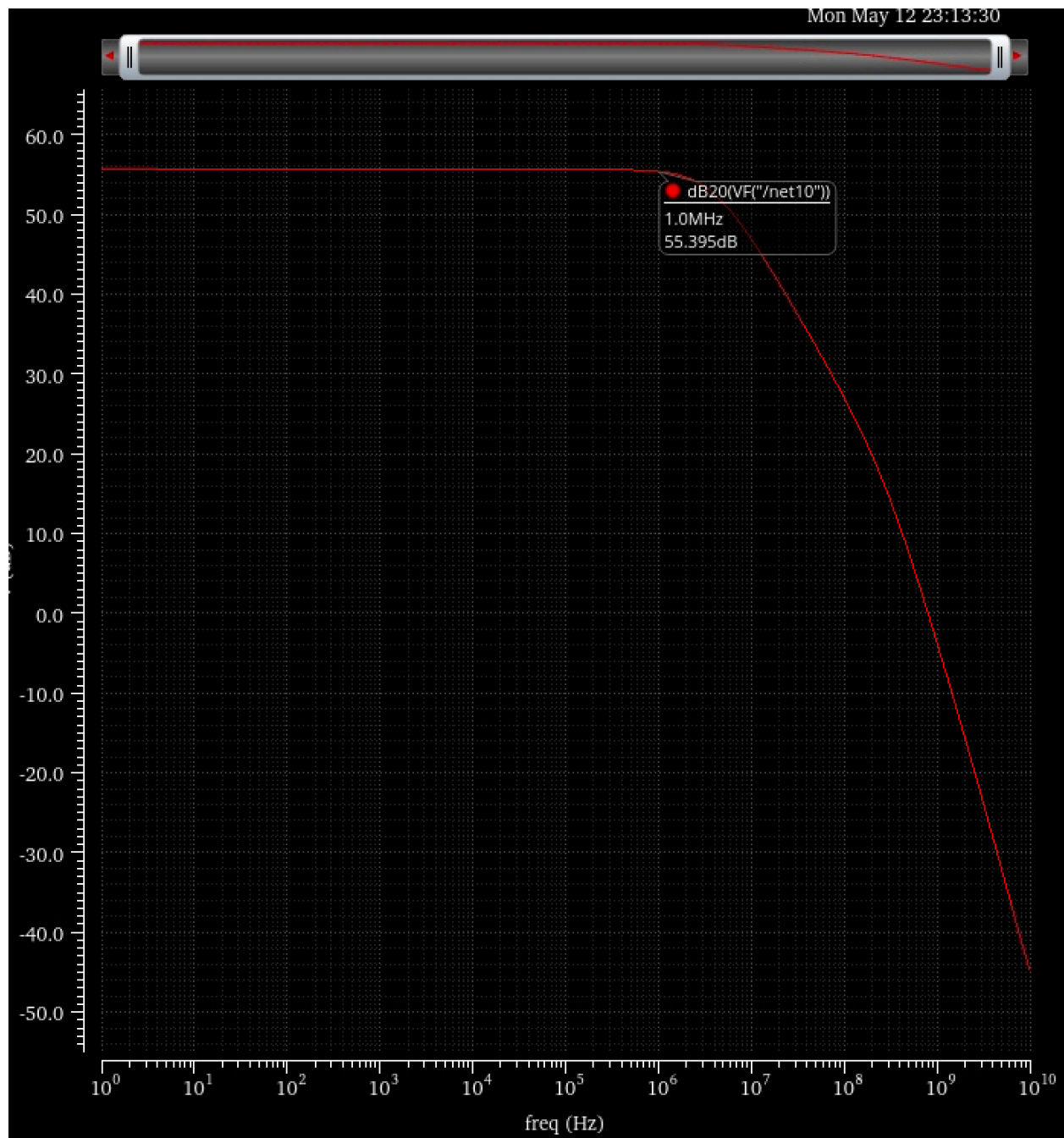
Amplifier Schematic



Amplifier Test Bench

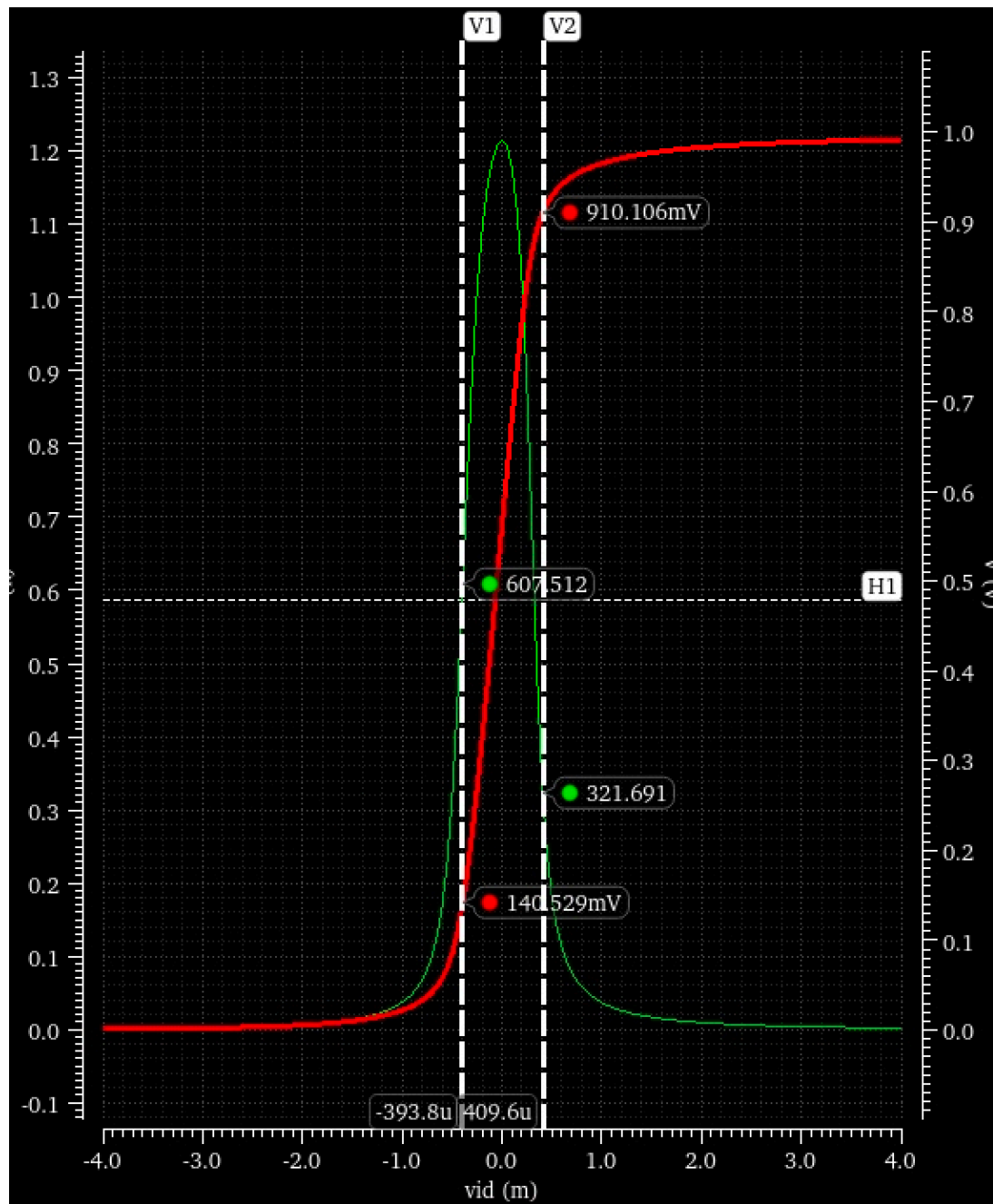


Amplifier Gain 55db to linear: 588.5



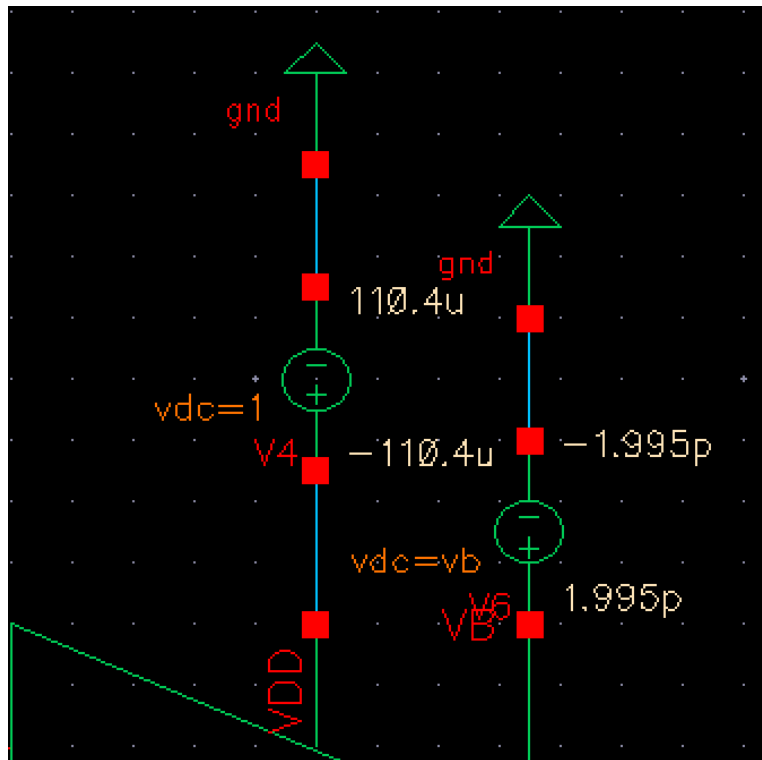
90 Percent of Gain = 529.2

Amplifier Swing: 910V - 140V = 770V



Power Usage:

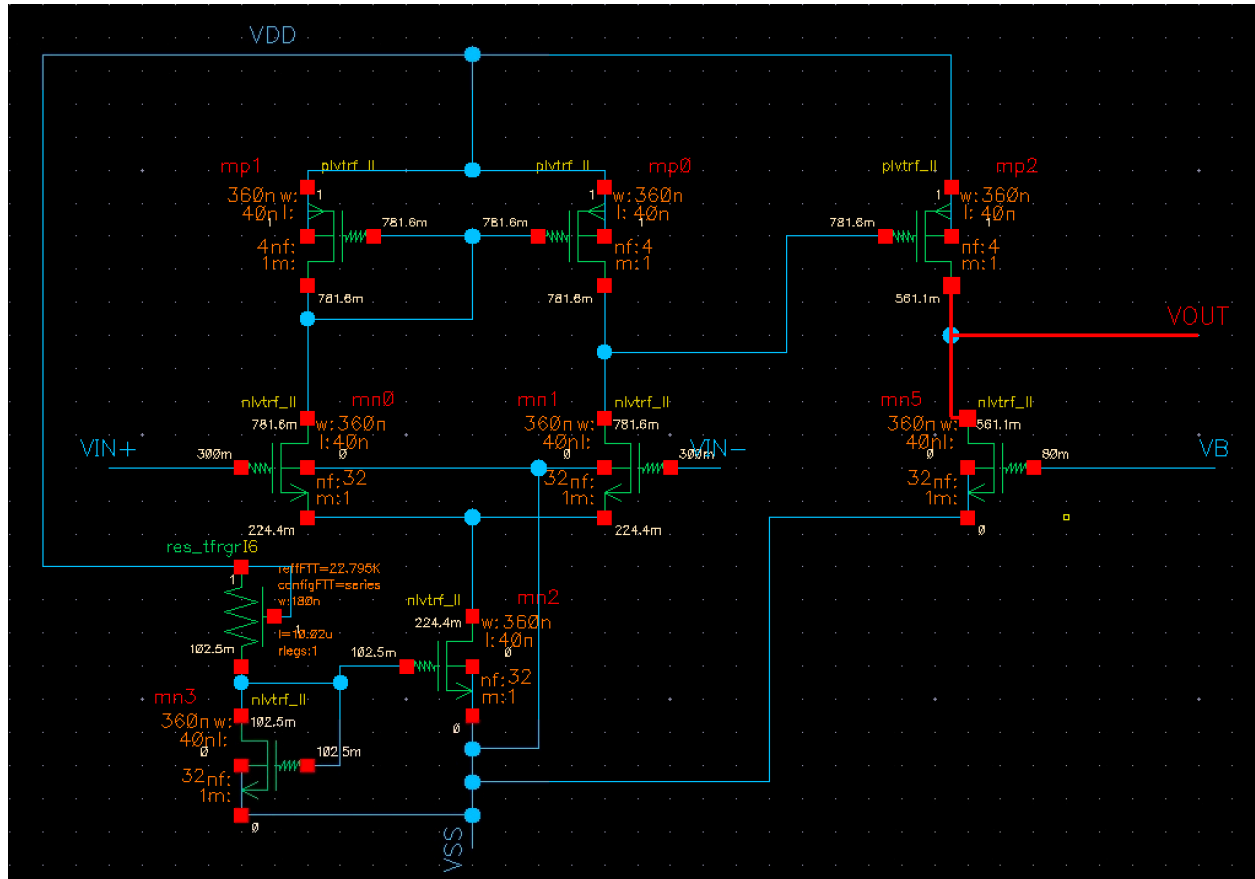
$$110.4\mu\text{W} + \sim 0 = 110\mu\text{W}$$



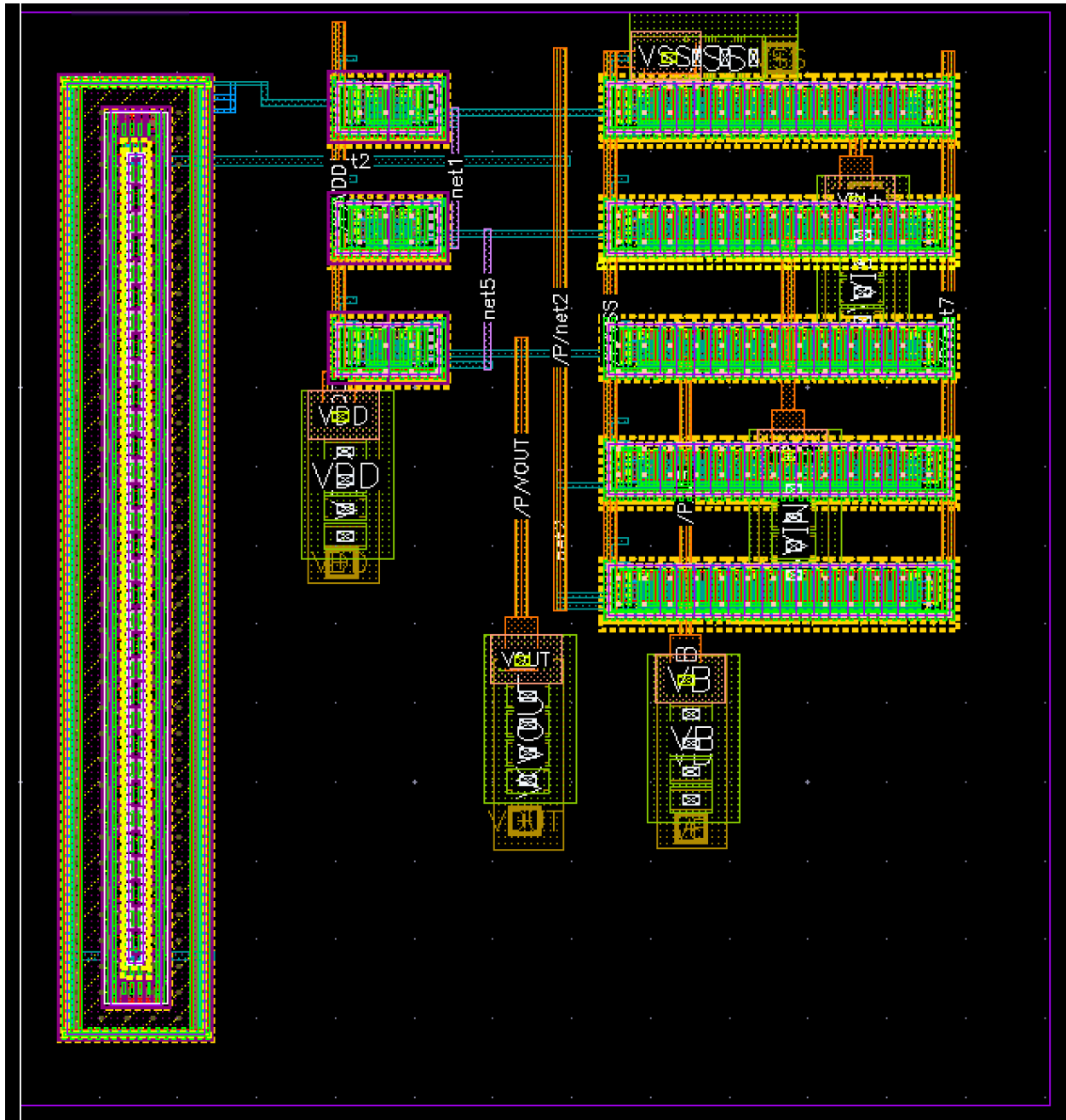
FOM:

$$55 \times 0.529 / 0.110 = 264.5$$

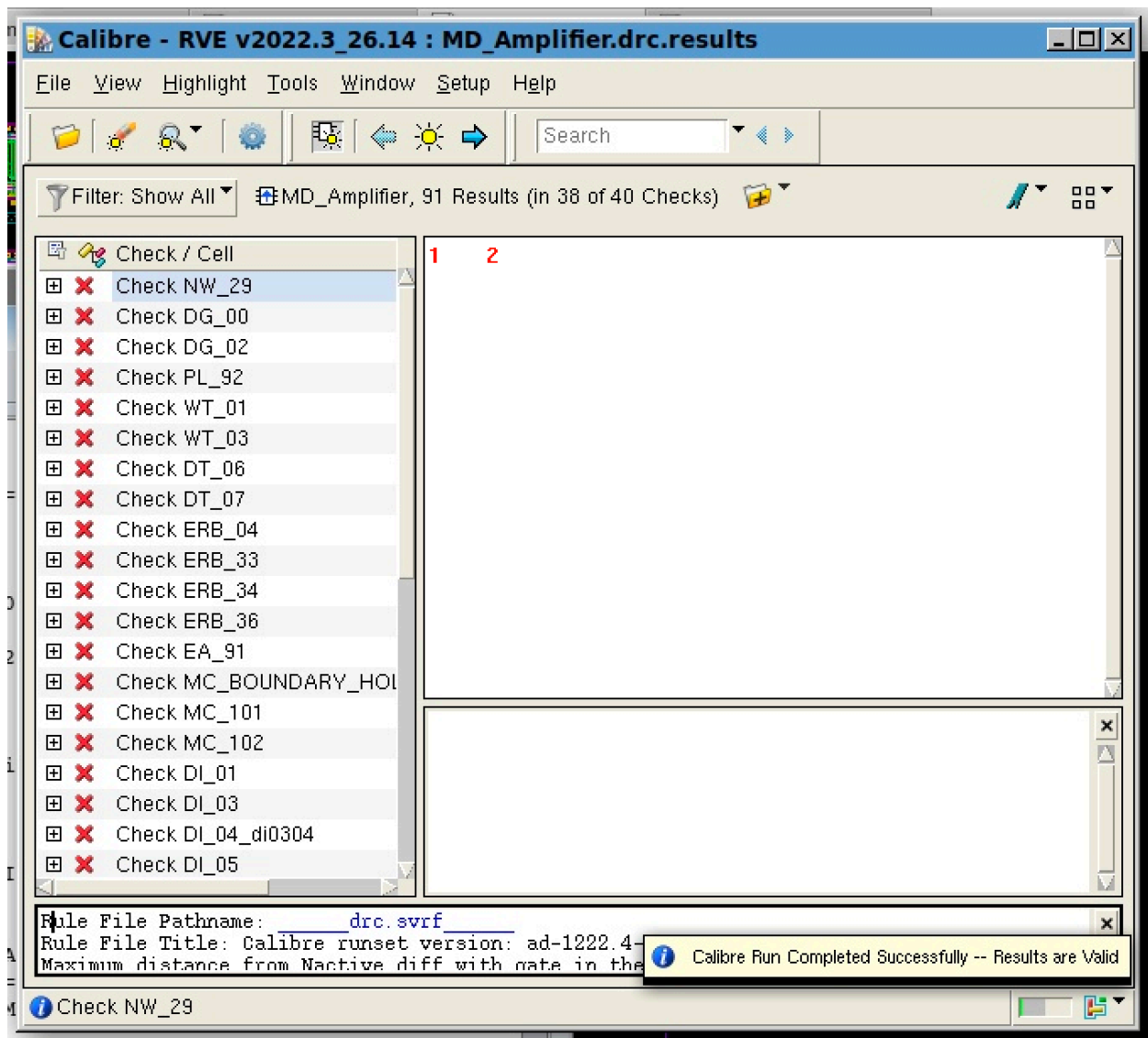
Amplifier Node Voltages



Amplifier Layout



Amplifier DRC



Amplifier LVS Clean

Calibre - RVE v2022.3_26.14 : svdb MD_Amplifier

File View Highlight Tools Window Setup Help

Search

Navigator

- Results
 - Extraction Results
 - Comparison Results
- ERC
 - ERC Results
 - ERC Summary
- Reports
 - Extraction Report
 - LVS Report
- Rules
 - Rules File
- View
 - Info
 - Finder
 - Schematics
- Setup
 - Options

Comparison Results

Layout Cell / Type	Source Cell	Nets	Instances	Ports
MD_Amplifier	MD_Amplifier	10L, 10S	9L, 9S	6L, 6S

Cell MD_Amplifier Summary (Clean)

CELL COMPARISON RESULTS (TOP LEVEL)

```
#####  
#          #  
# CORRECT  #  
#          #  
#####
```

LAYOUT CELL NAME: MD_Amplifier
SOURCE CELL NAME: MD_Amplifier

INITIAL NUMBERS OF OBJECTS

	Layout	Source	Component Type
Ports:	6	6	

Calibre - RVE v2022.3_26.14 : dejesus_62080_2025.drc.results

File View Highlight Tools Window Setup Help

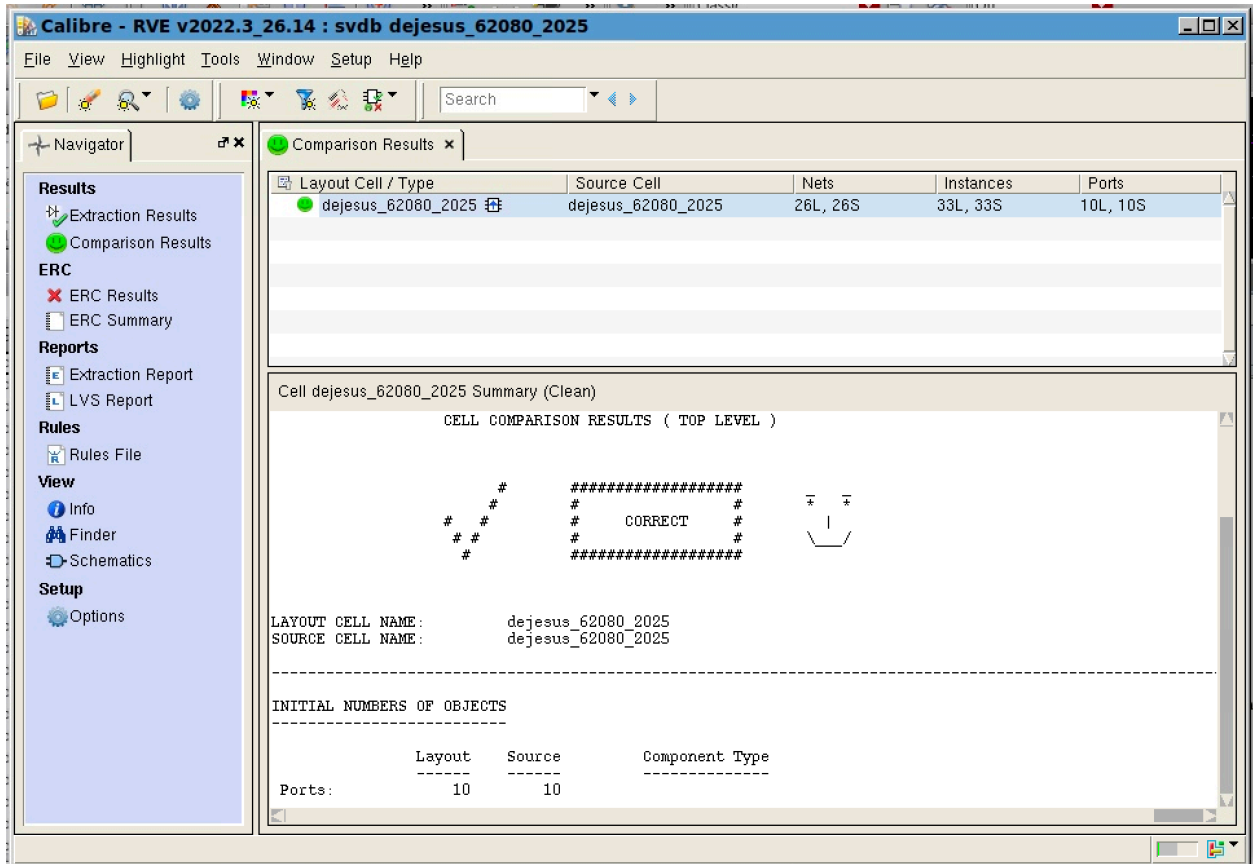
Filter: Show All dejesus_62080_2025, 64 Results (in 41 of 43 Checks)

Check / Cell	Results	Flat
Check NW_29	4	176
Check DG_00	10	281
Check DG_02	10	281
Check PL_92	1	1
Check WT_03	1	1
Check ERB_04	1	1
Check ERB_33	1	1
Check ERB_34	1	1
Check ERB_36	4	4
Check EA_91	1	1
Check MC_BOUNDARY_HOLE	1	1
Check MC_101	1	1
Check MC_102	1	1
Check DI_01	2	2
Check DI_03	1	1

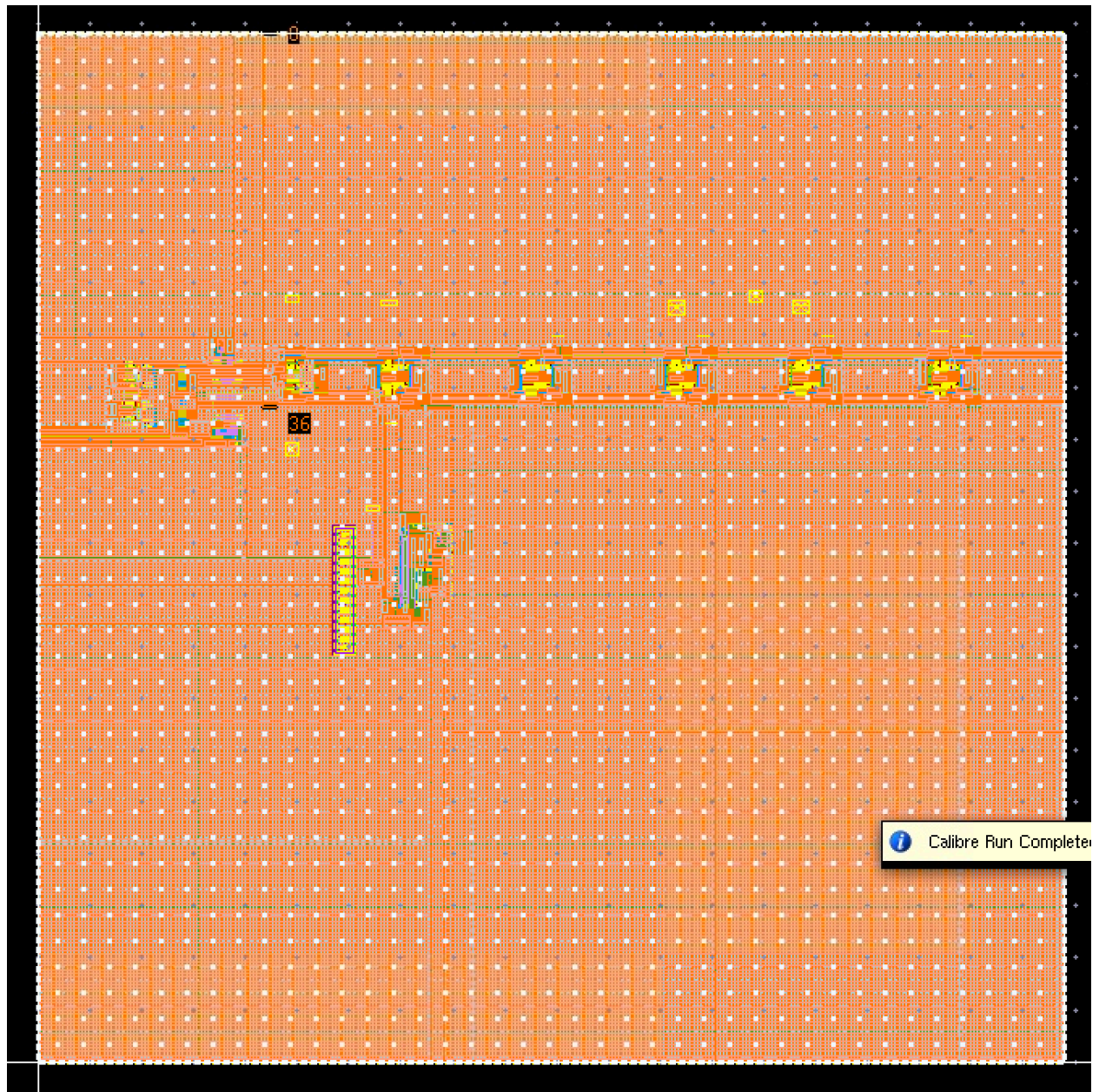
Rule File Pathname: drc.svrf
 Rule File Title: Calibre runset version: ad-1222.4-4.1.0.3/10
 Maximum distance from Nactive diff with gate in the pwell to a sub tap. (This check may require tap to have connection to bump(c4b) including for VSS nets or may require tap to have connection to port at top level). Note: For opt6 and opt7 ALP should be used instead of C4B for this check. = 39

Calibre Run Completed Successfully -- Results are Valid

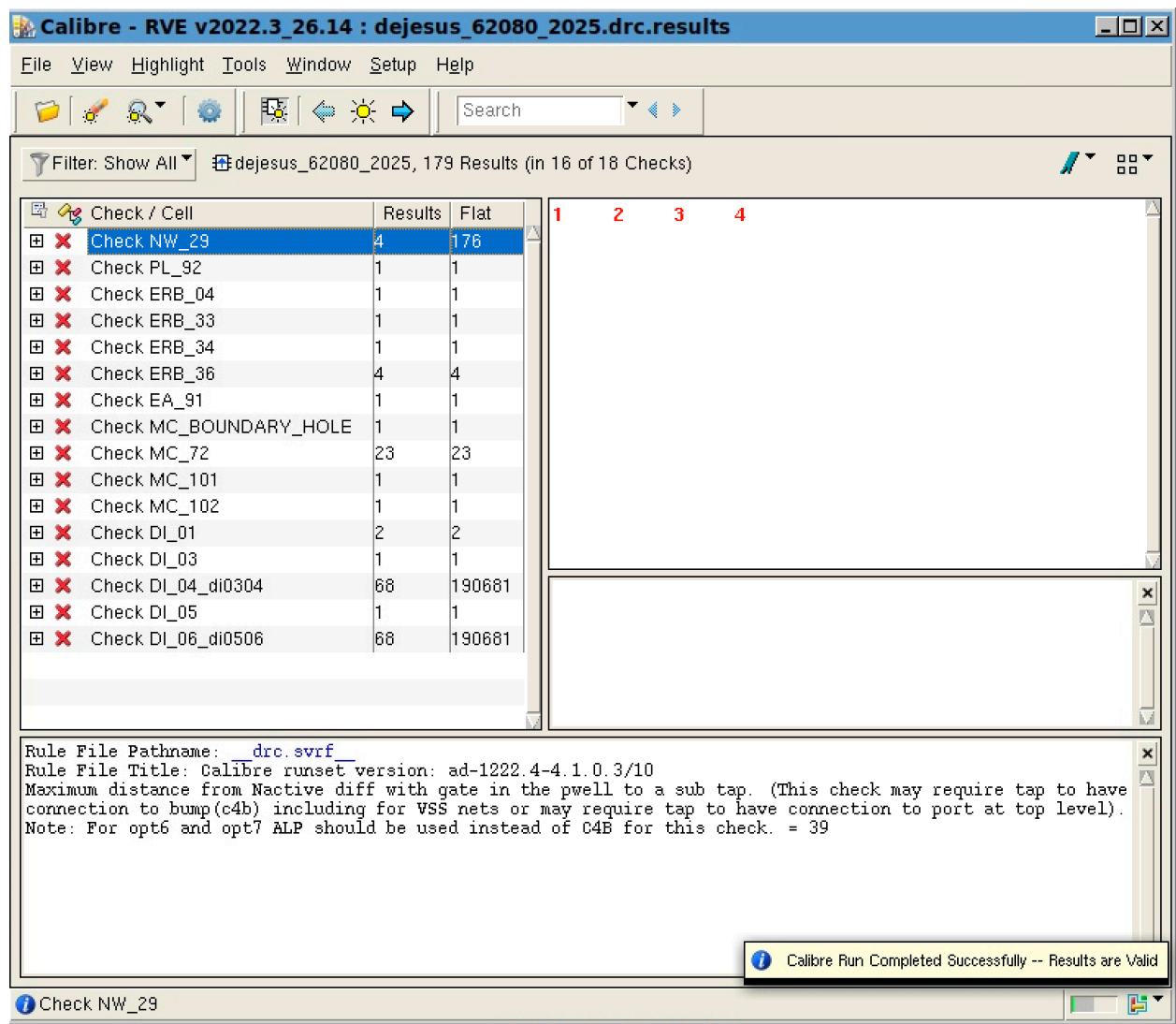
Check NW_29



Full System (Post Fill)



Full System DRC Post Fill



Full System LVS

Calibre - RVE v2022.3_26.14 : svdb dejesus_62080_2025

File View Highlight Tools Window Setup Help

Search

Navigator

Results

- Extraction Results
- Comparison Results

ERC

- ERC Results
- ERC Summary

Reports

- Extraction Report
- LVS Report

Rules

- Rules File

View

- Info
- Finder
- Schematics

Setup

- Options

Comparison Results

Layout Cell / Type	Source Cell	Nets	Instances	Ports
dejesus_62080_2025	dejesus_62080_2025	26L, 26S	33L, 33S	10L, 10S

Cell dejesus_62080_2025 Summary (Clean)

CELL COMPARISON RESULTS (TOP LEVEL)

```
# # # # # # # # # # # # # # # #
# # # # # # # # # # # # # # # #
# # # # # # # # # # # # # # # #
# # # # # # # # # # # # # # # #
# # # # # # # # # # # # # # # #
```

LAYOUT CELL NAME: dejesus_62080_2025
SOURCE CELL NAME: dejesus_62080_2025

INITIAL NUMBERS OF OBJECTS

	Layout	Source	Component Type
Ports:	10	10	
Nets:	189	26	*

